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Computer Organization

and Architecture

8th Edition

Chapter 2

Computer Evolution and

Performance

ENIAC - background

- Electronic Numerical Integrator And Computer
- Eckert and Mauchly
- University of Pennsylvania
- Trajectory tables for weapons
- Started 1943
- Finished 1946
 - Too late for war effort
- Used until 1955

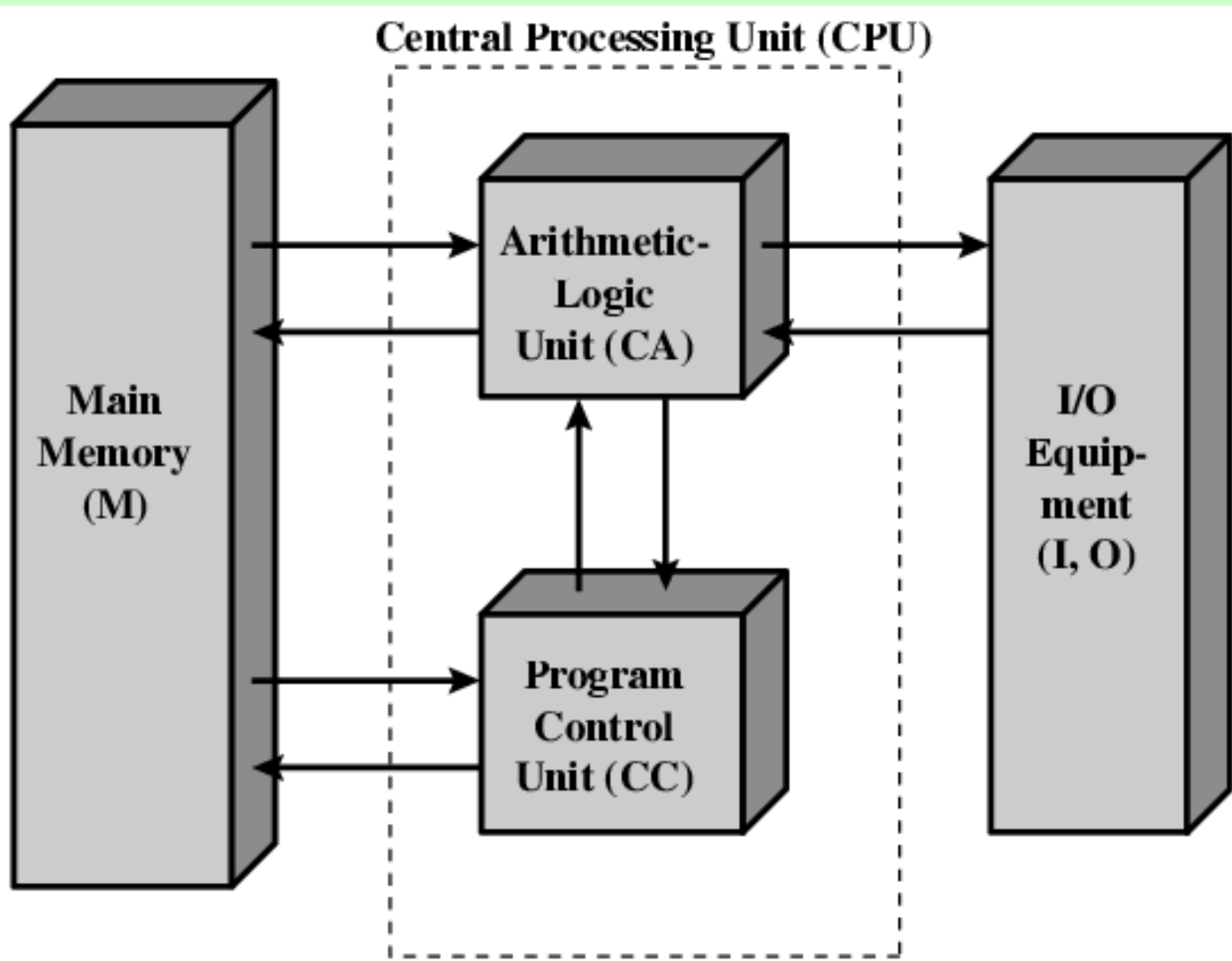
ENIAC - details

- Decimal (not binary)
- 20 accumulators of 10 digits
- Programmed manually by switches
- 18,000 vacuum tubes
- 30 tons
- 15,000 square feet
- 140 kW power consumption
- 5,000 additions per second

von Neumann/Turing

- Stored Program concept
- Main memory storing programs and data
- ALU operating on binary data
- Control unit interpreting instructions from memory and executing
- Input and output equipment operated by control unit
- Princeton Institute for Advanced Studies
 - IAS
- Completed 1952

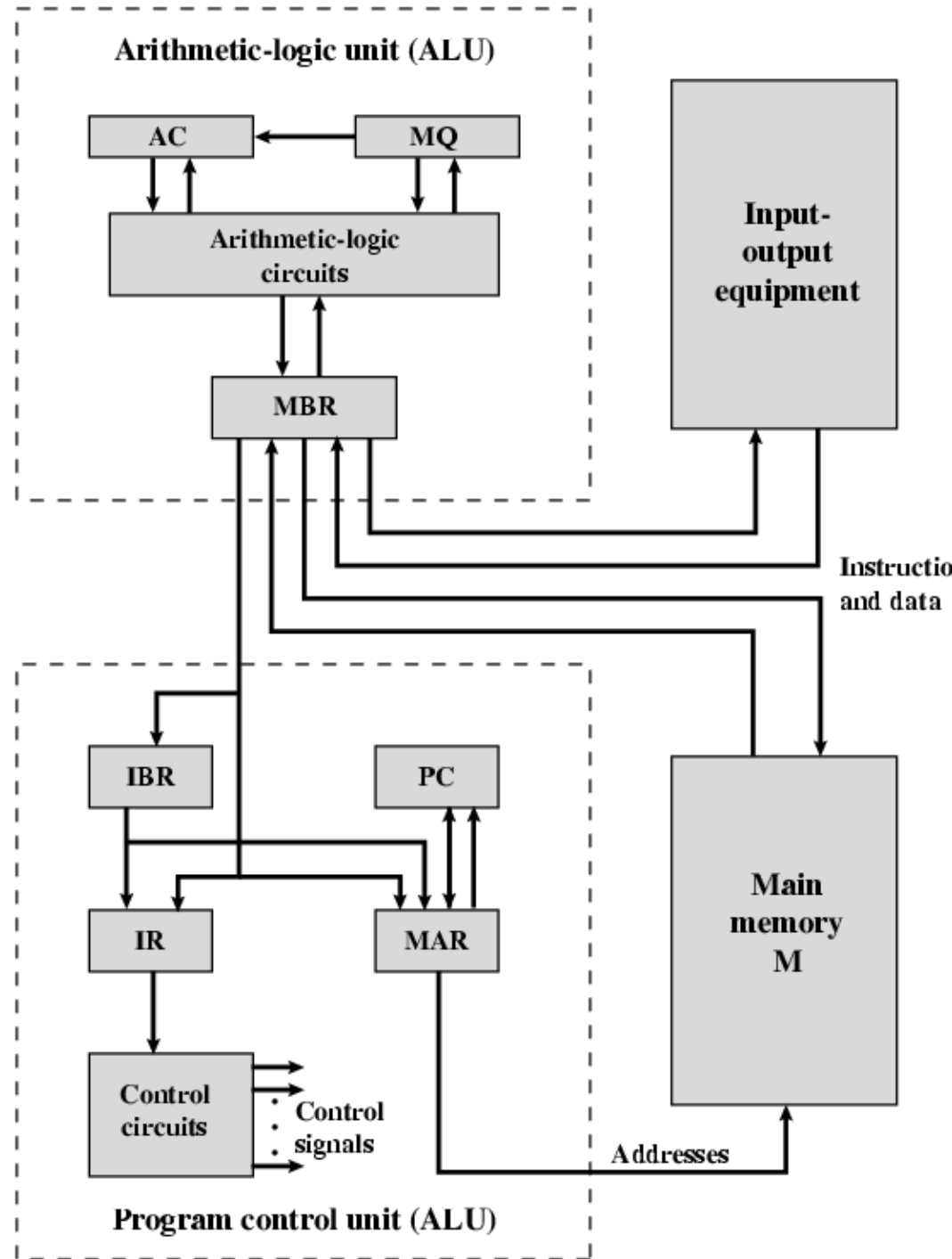
Structure of von Neumann machine



IAS - details

- 1000 x 40 bit words
 - Binary number
 - 2 x 20 bit instructions
- Set of registers (storage in CPU)
 - Memory Buffer Register
 - Memory Address Register
 - Instruction Register
 - Instruction Buffer Register
 - Program Counter
 - Accumulator
 - Multiplier Quotient

Structure of IAS – detail



Commercial Computers

- 1947 - Eckert-Mauchly Computer Corporation
- UNIVAC I (Universal Automatic Computer)
- US Bureau of Census 1950 calculations
- Became part of Sperry-Rand Corporation
- Late 1950s - UNIVAC II
 - Faster
 - More memory

IBM

- Punched-card processing equipment
- 1953 - the 701
 - IBM's first stored program computer
 - Scientific calculations
- 1955 - the 702
 - Business applications
- Lead to 700/7000 series

Transistors

- Replaced vacuum tubes
- Smaller
- Cheaper
- Less heat dissipation
- Solid State device
- Made from Silicon (Sand)
- Invented 1947 at Bell Labs
- William Shockley et al.

Transistor Based Computers

- Second generation machines
- NCR & RCA produced small transistor machines
- IBM 7000
- DEC - 1957
 - Produced PDP-1

Microelectronics

- Literally - “small electronics”
- A computer is made up of gates, memory cells and interconnections
- These can be manufactured on a semiconductor
- e.g. silicon wafer

Generations of Computer

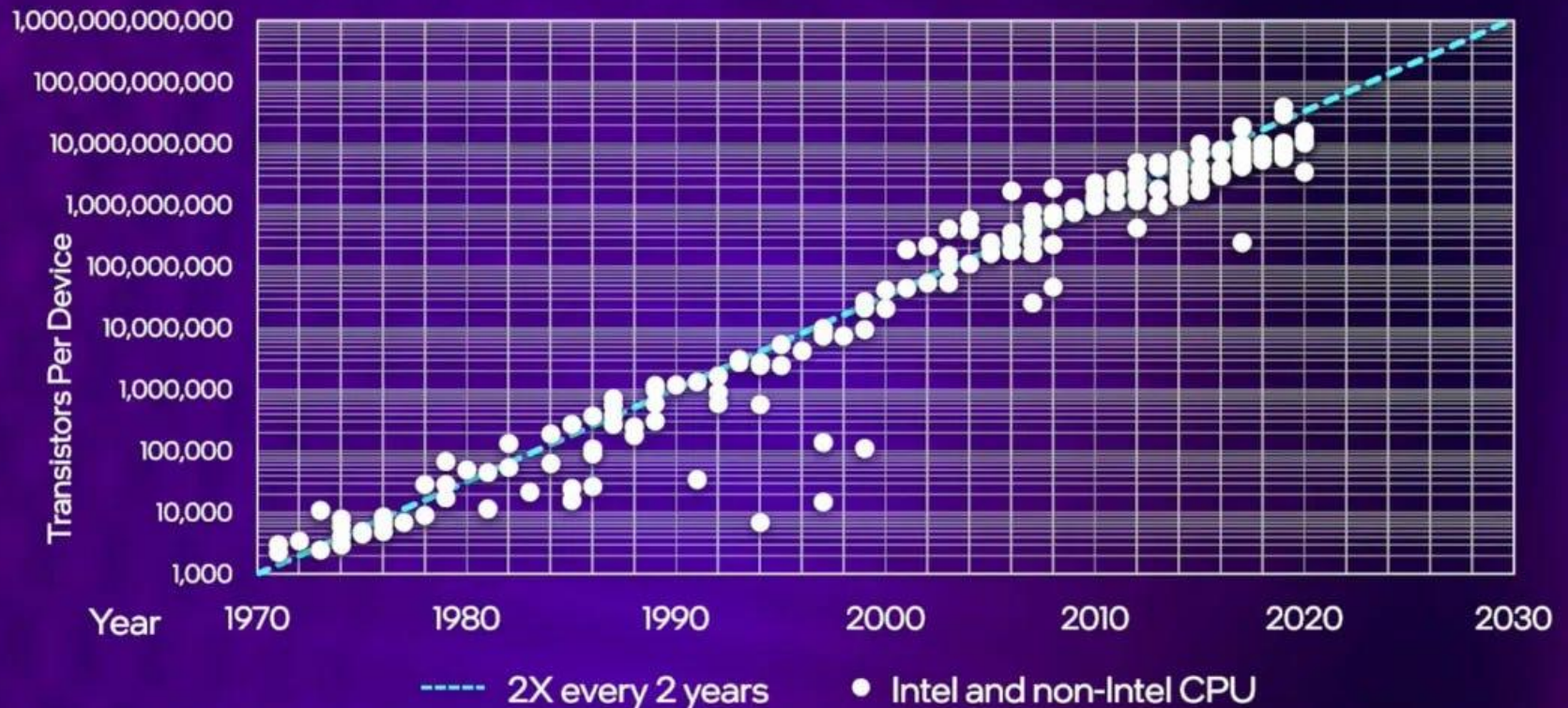
- Vacuum tube - 1946-1957
- Transistor - 1958-1964
- Small scale integration - 1965 on
 - Up to 100 devices on a chip
- Medium scale integration - to 1971
 - 100-3,000 devices on a chip
- Large scale integration - 1971-1977
 - 3,000 - 100,000 devices on a chip
- Very large scale integration - 1978 -1991
 - 100,000 - 100,000,000 devices on a chip
- Ultra large scale integration – 1991 -
 - Over 100,000,000 devices on a chip

Moore's Law

- Increased density of components on chip
- Gordon Moore – co-founder of Intel
- Number of transistors on a chip will double every year
- Since 1970's development has slowed a little
 - Number of transistors doubles every 18 months
- Cost of a chip has remained almost unchanged
- Higher packing density means shorter electrical paths, giving higher performance
- Smaller size gives increased flexibility
- Reduced power and cooling requirements
- Fewer interconnections increases reliability

Growth in CPU Transistor Count

Moore's Law is alive and well



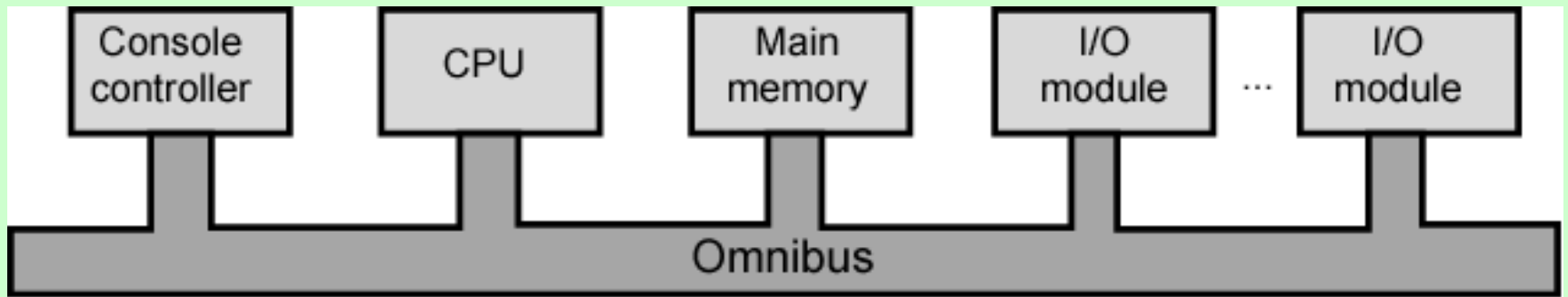
IBM 360 series

- 1964
- Replaced (& not compatible with) 7000 series
- First planned “family” of computers
 - Similar or identical instruction sets
 - Similar or identical O/S
 - Increasing speed
 - Increasing number of I/O ports (i.e. more terminals)
 - Increased memory size
 - Increased cost
- Multiplexed switch structure

DEC PDP-8

- 1964
- First minicomputer (after miniskirt!)
- Did not need air conditioned room
- Small enough to sit on a lab bench
- \$16,000
 - \$100k+ for IBM 360
- Embedded applications & OEM
- BUS STRUCTURE

DEC - PDP-8 Bus Structure



Semiconductor Memory

- 1970
- Fairchild
- Size of a single core
 - i.e. 1 bit of magnetic core storage
- Holds 256 bits
- Non-destructive read
- Much faster than core
- Capacity approximately doubles each year

Intel

- 1971 - 4004
 - First microprocessor
 - All CPU components on a single chip
 - 4 bit
- Followed in 1972 by 8008
 - 8 bit
 - Both designed for specific applications
- 1974 - 8080
 - Intel's first general purpose microprocessor

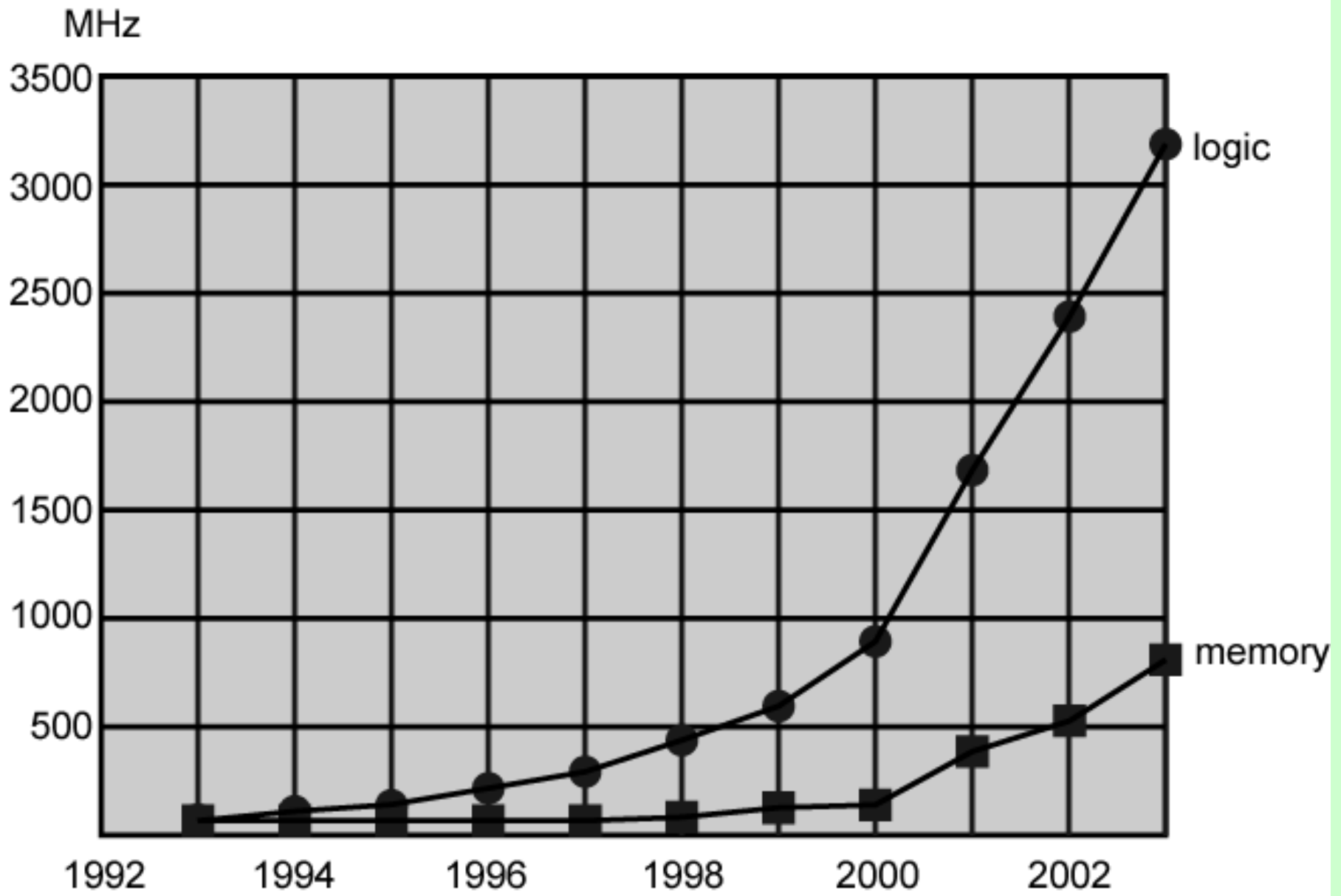
Speeding it up

- Pipelining
- On board cache
- On board L1 & L2 cache
- Branch prediction
- Data flow analysis
- Speculative execution

Performance Balance

- Processor speed increased
- Memory capacity increased
- Memory speed lags behind processor speed

Login and Memory Performance Gap



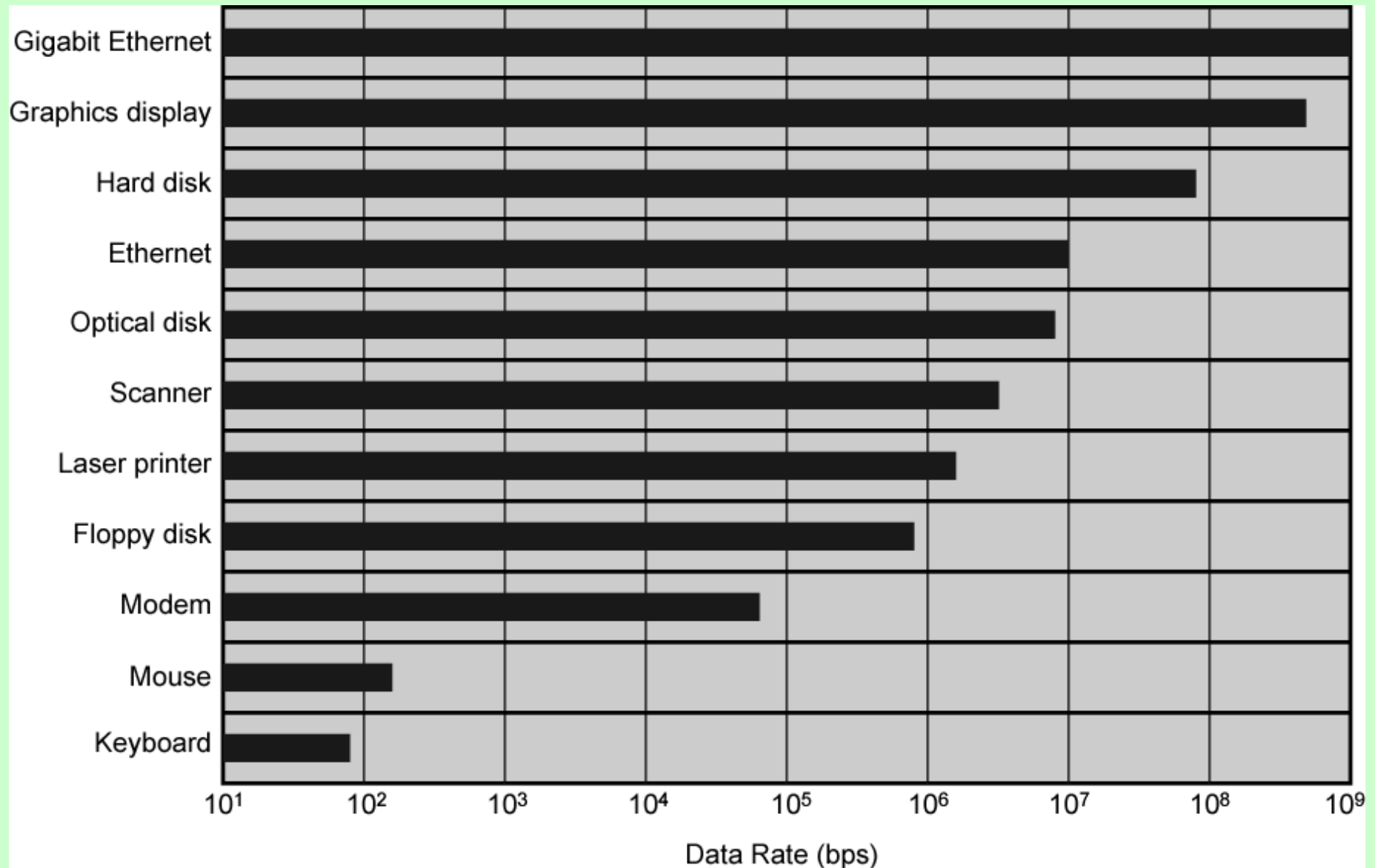
Solutions

- Increase number of bits retrieved at one time
 - Make DRAM “wider” rather than “deeper”
- Change DRAM interface
 - Cache
- Reduce frequency of memory access
 - More complex cache and cache on chip
- Increase interconnection bandwidth
 - High speed buses
 - Hierarchy of buses

I/O Devices

- Peripherals with intensive I/O demands
- Large data throughput demands
- Processors can handle this
- Problem moving data
- Solutions:
 - Caching
 - Buffering
 - Higher-speed interconnection buses
 - More elaborate bus structures
 - Multiple-processor configurations

Typical I/O Device Data Rates



Key is Balance

- Processor components
- Main memory
- I/O devices
- Interconnection structures

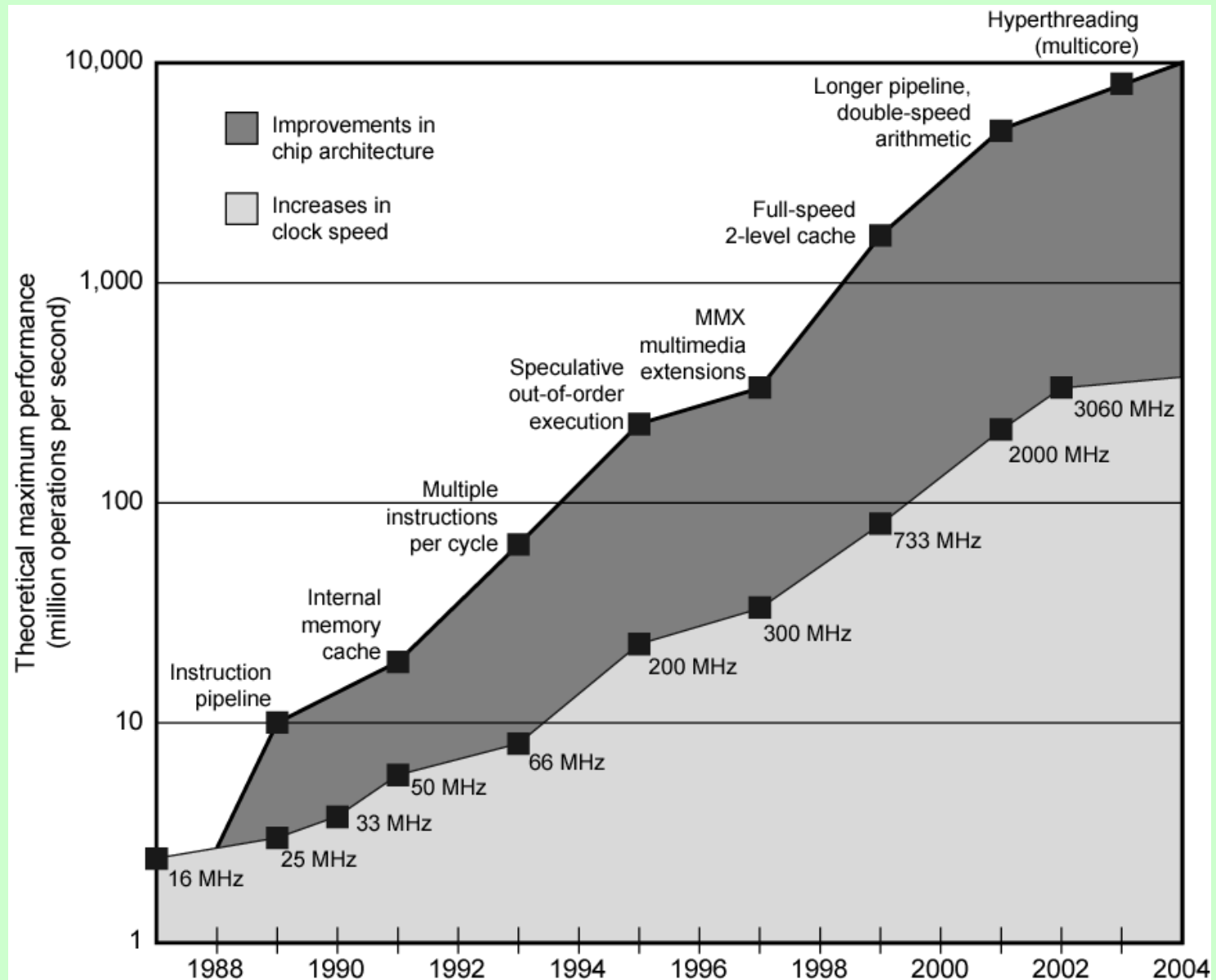
Improvements in Chip Organization and Architecture

- Increase hardware speed of processor
 - Fundamentally due to shrinking logic gate size
 - More gates, packed more tightly, increasing clock rate
 - Propagation time for signals reduced
- Increase size and speed of caches
 - Dedicating part of processor chip
 - Cache access times drop significantly
- Change processor organization and architecture
 - Increase effective speed of execution
 - Parallelism

Problems with Clock Speed and Login Density

- Power
 - Power density increases with density of logic and clock speed
 - Dissipating heat
- RC delay
 - Speed at which electrons flow limited by resistance and capacitance of metal wires connecting them
 - Delay increases as RC product increases
 - Wire interconnects thinner, increasing resistance
 - Wires closer together, increasing capacitance
- Memory latency
 - Memory speeds lag processor speeds
- Solution:
 - More emphasis on organizational and architectural approaches

Intel Microprocessor Performance



Increased Cache Capacity

- Typically two or three levels of cache between processor and main memory
- Chip density increased
 - More cache memory on chip
 - Faster cache access
- Pentium chip devoted about 10% of chip area to cache
- Pentium 4 devotes about 50%

More Complex Execution Logic

- Enable parallel execution of instructions
- Pipeline works like assembly line
 - Different stages of execution of different instructions at same time along pipeline
- Superscalar allows multiple pipelines within single processor
 - Instructions that do not depend on one another can be executed in parallel

Diminishing Returns

- Internal organization of processors complex
 - Can get a great deal of parallelism
 - Further significant increases likely to be relatively modest
- Benefits from cache are reaching limit
- Increasing clock rate runs into power dissipation problem
 - Some fundamental physical limits are being reached

New Approach – Multiple Cores

- Multiple processors on single chip
 - Large shared cache
- Within a processor, increase in performance proportional to square root of increase in complexity
- If software can use multiple processors, doubling number of processors almost doubles performance
- So, use two simpler processors on the chip rather than one more complex processor
- With two processors, larger caches are justified
 - Power consumption of memory logic less than processing logic

x86 Evolution (1)

- 8080
 - first general purpose microprocessor
 - 8 bit data path
 - Used in first personal computer – Altair
- 8086 – 5MHz – 29,000 transistors
 - much more powerful
 - 16 bit
 - instruction cache, prefetch few instructions
 - 8088 (8 bit external bus) used in first IBM PC
- 80286
 - 16 Mbyte memory addressable
 - up from 1Mb
- 80386
 - 32 bit
 - Support for multitasking
- 80486
 - sophisticated powerful cache and instruction pipelining
 - built in maths co-processor

x86 Evolution (2)

- Pentium
 - Superscalar
 - Multiple instructions executed in parallel
- Pentium Pro
 - Increased superscalar organization
 - Aggressive register renaming
 - branch prediction
 - data flow analysis
 - speculative execution
- Pentium II
 - MMX technology
 - graphics, video & audio processing
- Pentium III
 - Additional floating point instructions for 3D graphics

x86 Evolution (3)

- Pentium 4
 - Note Arabic rather than Roman numerals
 - Further floating point and multimedia enhancements
- Core
 - First x86 with dual core
- Core 2
 - 64 bit architecture
- Core 2 Quad – 3GHz – 820 million transistors
 - Four processors on chip
- x86 architecture dominant outside embedded systems
- Organization and technology changed dramatically
- Instruction set architecture evolved with backwards compatibility
- ~1 instruction per month added
- 500 instructions available
- See Intel web pages for detailed information on processors

Embedded Systems

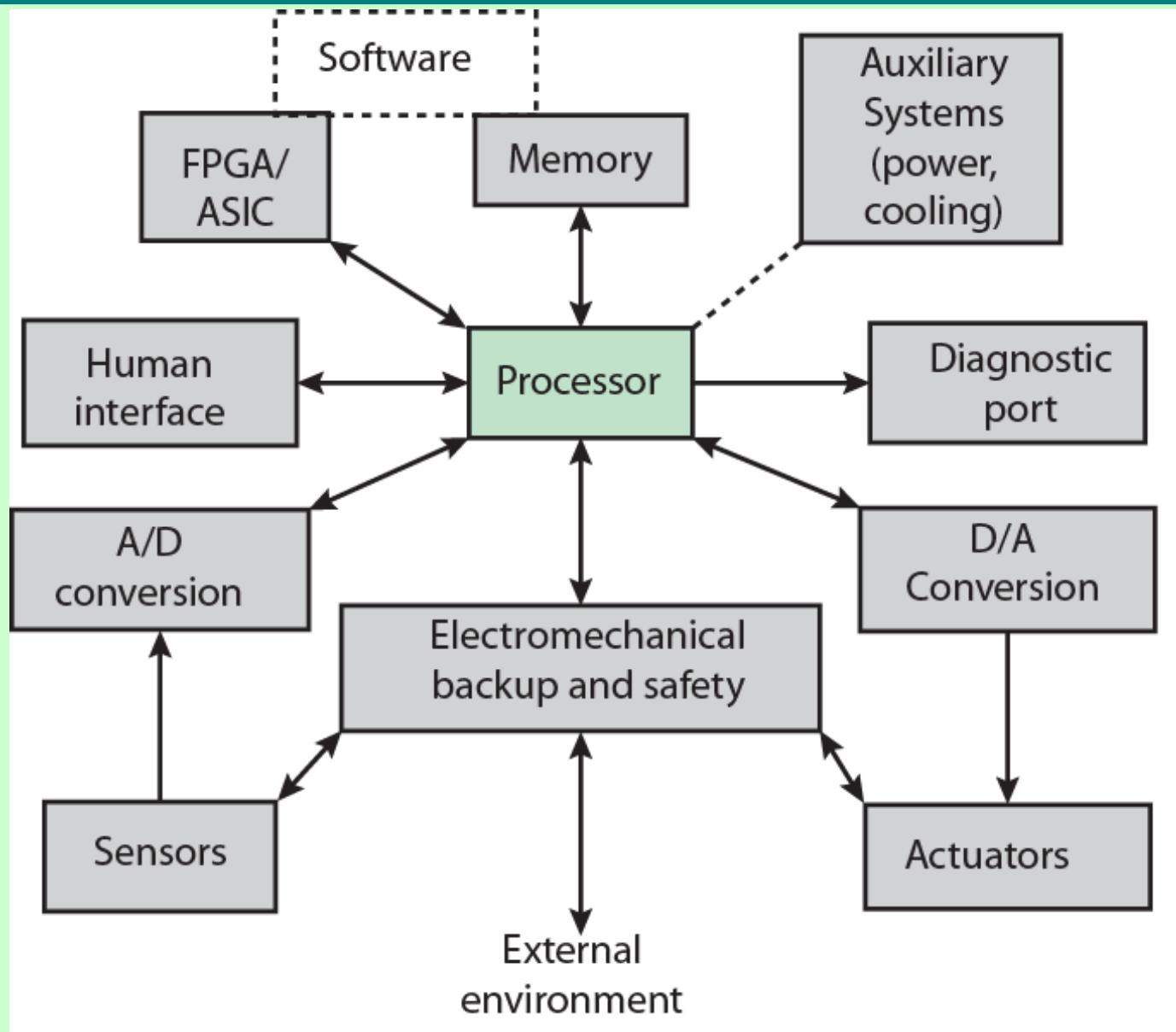
ARM

- ARM evolved from RISC design
- Used mainly in embedded systems
 - Used within product
 - Not general purpose computer
 - Dedicated function
 - E.g. Anti-lock brakes in car

Embedded Systems Requirements

- Different sizes
 - Different constraints, optimization, reuse
- Different requirements
 - Safety, reliability, real-time, flexibility, legislation
 - Lifespan
 - Environmental conditions
 - Static v dynamic loads
 - Slow to fast speeds
 - Computation v I/O intensive
 - Discrete event v continuous dynamics

Possible Organization of an Embedded System



ARM Evolution

- Designed by ARM Inc., Cambridge, England
- Licensed to manufacturers
- High speed, small die, low power consumption
- PDAs, hand held games, phones
 - E.g. iPod, iPhone
- Acorn produced ARM1 & ARM2 in 1985 and ARM3 in 1989
- Acorn, VLSI and Apple Computer founded ARM Ltd.

ARM Systems Categories

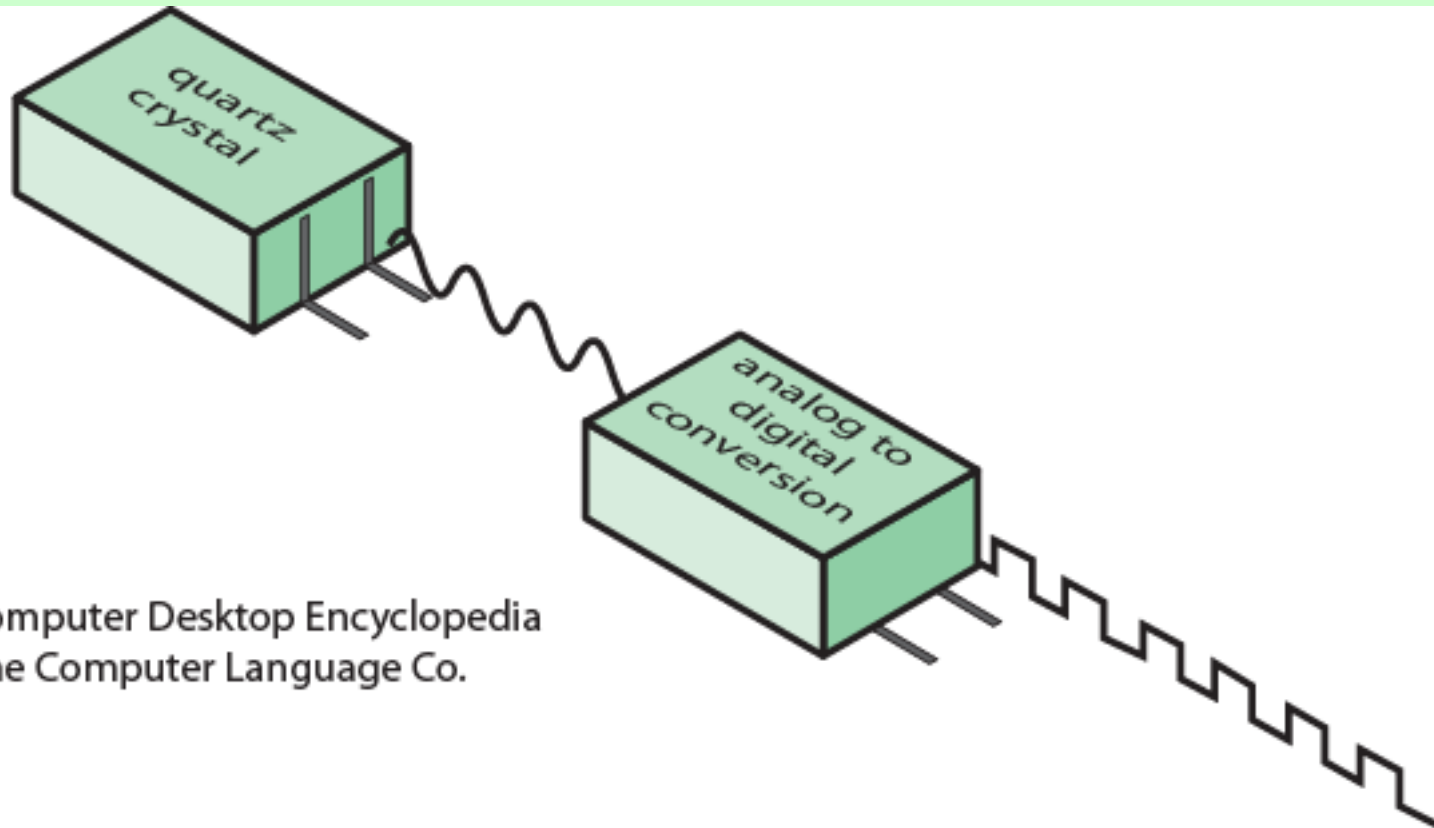
- Embedded real time
- Application platform
 - Linux, Palm OS, Symbian OS, Windows mobile
- Secure applications

Performance Assessment

Clock Speed

- Key parameters
 - Performance, cost, size, security, reliability, power consumption
- System clock speed
 - In Hz or multiples of
 - Clock rate, clock cycle, clock tick, cycle time
- Signals in CPU take time to settle down to 1 or 0
- Signals may change at different speeds
- Operations need to be synchronised
- Instruction execution in discrete steps
 - Fetch, decode, load and store, arithmetic or logical
 - Usually require multiple clock cycles per instruction
- Pipelining gives simultaneous execution of instructions
- So, clock speed is not the whole story

System Clock



From Computer Desktop Encyclopedia
1998, The Computer Language Co.

Instruction Execution Rate

- Millions of instructions per second (MIPS)
- Millions of floating point instructions per second (MFLOPS)
- Heavily dependent on instruction set, compiler design, processor implementation, cache & memory hierarchy

Benchmarks

- Programs designed to test performance
- Written in high level language
 - Portable
- Represents style of task
 - Systems, numerical, commercial
- Easily measured
- Widely distributed
- E.g. System Performance Evaluation Corporation (SPEC)
 - CPU2006 for computation bound
 - 17 floating point programs in C, C++, Fortran
 - 12 integer programs in C, C++
 - 3 million lines of code
 - Speed and rate metrics
 - Single task and throughput

SPEC Speed Metric

- Single task
- Base runtime defined for each benchmark using reference machine
- Results are reported as ratio of reference time to system run time
 - T_{ref_i} execution time for benchmark i on reference machine
 - T_{sut_i} execution time of benchmark i on test system

$$r_i = \frac{T_{ref_i}}{T_{sut_i}}$$

- Overall performance calculated by averaging ratios for all 12 integer benchmarks
 - Use geometric mean
 - Appropriate for normalized numbers such as ratios

$$r_G = \left(\prod_{i=1}^n r_i \right)^{1/n}$$

SPEC Rate Metric

- Measures throughput or rate of a machine carrying out a number of tasks
- Multiple copies of benchmarks run simultaneously
 - Typically, same as number of processors
- Ratio is calculated as follows:
 - T_{ref_i} reference execution time for benchmark i
 - N number of copies run simultaneously
 - T_{sut_i} elapsed time from start of execution of program on all N processors until completion of all copies of program
 - Again, a geometric mean is calculated

$$r_i = \frac{N \times T_{ref_i}}{T_{sut_i}}$$

Amdahl's Law

- Gene Amdahl [AMDA67]
- Potential speed up of program using multiple processors
- Concluded that:
 - Code needs to be parallelizable
 - Speed up is bound, giving diminishing returns for more processors
- Task dependent
 - Servers gain by maintaining multiple connections on multiple processors
 - Databases can be split into parallel tasks

Amdahl's Law Formula

- For program running on single processor
 - Fraction f of code infinitely parallelizable with no scheduling overhead
 - Fraction $(1-f)$ of code inherently serial
 - T is total execution time for program on single processor
 - N is number of processors that fully exploit parallel portions of code

$$\text{Speedup} = \frac{\text{time to execute program on a single processor}}{\text{time to execute program on } N \text{ parallel processors}} = \frac{T(1-f) + Tf}{T(1-f) + \frac{Tf}{N}} = \frac{1}{(1-f) + \frac{f}{N}}$$

- Conclusions
 - f small, parallel processors has little effect
 - $N \rightarrow \infty$, speedup bound by $1/(1-f)$
 - Diminishing returns for using more processors

Internet Resources

- <http://www.intel.com/>
 - Search for the Intel Museum
- <http://www.ibm.com>
- <http://www.dec.com>
- Charles Babbage Institute
- PowerPC
- Intel Developer Home

References

- AMDA67 Amdahl, G. "Validity of the Single-Processor Approach to Achieving Large-Scale Computing Capability", *Proceedings of the AFIPS Conference*, 1967.